FAST VIVA VOCE & ORAL EXAMINATION CHEAT SHEET

25 Short Questions & Answers: The Whole Project in Plain English

Short, Punchy, High-Scoring Answers That Are Easy to Memorize and Speak Aloud

Target Audience:	Students facing final year project viva, external academic defense, or technical interviews.
Answer Format:	1 to 2 spoken sentences per question. Zero unnecessary fluff. High technical impact.
Topics Covered:	Project Overview, Dual-Portal Architecture, Gemini AI, Supabase PostgreSQL, Scoring & Modules.
How to Practice:	Read each answer out loud 2 to 3 times before your meeting to build natural speaking confidence.

Golden Rule for Your Exam: Keep your answers concise! State the direct answer first, mention the exact tool or technology used, and stop. If the examiner wants more depth, they will ask. This cheat sheet gives you the exact high-scoring response for every question.

1. General Project & Architecture Questions

Q1: What is CareerForge AI?

Spoken Answer: CareerForge AI is a career guidance and placement preparation platform for engineering colleges. It guides students from career discovery to interview practice, and gives faculty a live dashboard to monitor readiness and endorse certificates.

Q2: What major problem does your project solve?

Spoken Answer: It bridges the gap between academic college syllabus and real-world hiring requirements, helping students discover their ideal engineering domain and practice with industry-grade AI interviews.

Q3: Who are the two end users of this platform?

Spoken Answer: Engineering students (who use the Student Portal at index.html) and college faculty/placement officers (who use the Admin Portal at admin.html).

Q4: What is the dual-portal architecture?

Spoken Answer: It means the application has two separate interfaces: a learning and practice portal for students, and an administration and governance portal for faculty, both synchronized in real time via a shared Supabase PostgreSQL database.

Q5: Why did you use Vanilla JavaScript instead of React or Angular?

Spoken Answer: Vanilla JavaScript eliminates heavy framework dependencies and build steps, loads instantly with sub-millisecond DOM speeds, and demonstrates our core mastery of fundamental web standards like hash routing, async event loops, and HTML5 Canvas.

Q6: How many modules are in the project?

Spoken Answer: There are **8 core modules**: 7 modules for the student journey (Auth, Career Discovery, Workstation Visualizer, 3D Resume Builder, Technical & HR Interviews, AI Mock Interview, Certification) and 1 module for Faculty Administration.

2. Artificial Intelligence (Google Gemini 3.6 Flash)

Q7: Which AI model are you using, and why?

Spoken Answer: We use **Google Gemini 3.6 Flash** (models/gemini-3.6-flash) because it offers sub-second response times, strict JSON output compliance, and uses the lowest API credits.

Q8: How does the Technical Interview work with AI?

Spoken Answer: When a student starts, Gemini acts as a **Principal Engineer** and dynamically generates 5 real-world technical MCQs tailored to their domain, complete with 4 options, the correct answer, and an in-depth technical explanation.

Q9: How does the HR Interview work with AI?

Spoken Answer: Gemini acts as a **Senior HR Director** and dynamically generates 5 behavioral and situational questions (conflict resolution, ethics, deadlines) along with the recruiter reasoning behind the best answer.

Q10: How does the AI Mock Interview work?

Spoken Answer: The student writes full paragraph answers to interview questions. Gemini evaluates the response and returns an honest 4-tier score breakdown: Relevance, Communication, Technical Depth, and Structure, plus coaching tips.

Q11: How do you prevent students from cheating the Mock Interview with gibberish?

Spoken Answer: We use a dual-layer filter. A client-side function checks vowel ratios (below 12% is flagged as keyboard mash), and Gemini's prompt strictly commands a **0% score** across all metrics for nonsense like 'sfdfdsfsnfbsjdfsdjf'.

3. Cloud Database & Backend (Supabase PostgreSQL)

Q12: Which database are you using, and where is it hosted?

Spoken Answer: We use **PostgreSQL 15** hosted in the cloud through **Supabase**.

Q13: What are the two main tables in Supabase?

Spoken Answer: auth.users for secure authentication credentials (email & password hash), and public.student_states for storing the student's actual scores and progress data.

Q14: Why is 'Display Name' empty in the Supabase Auth dashboard?

Spoken Answer: Because auth.users strictly handles authentication. Full student profile details (Name, Roll Number, Department, Year) are collected on the Profile Setup screen and saved in our student_states table.

Q15: Why do you store student progress as a JSONB column?

Spoken Answer: JSONB allows atomic single-trip reads and writes for our Single Page App without complex multi-table SQL joins, and allows us to add new badges or scores with zero database migration headaches.

Q16: How do the Student Portal and Admin Portal stay synchronized?

Spoken Answer: Every student action is automatically saved to the Supabase student_states table. The Admin portal queries this table in real-time, displaying live student records with zero dummy data.

Q17: What happens if a student loses internet while practicing?

Spoken Answer: We use dual-layer caching. The app saves to browser localStorage first so the interface never freezes, and then asynchronously syncs to Supabase once internet reconnects.

4. Core Features & Scoring System

Q18: How does the Career Discovery Assessment work?

Spoken Answer: A student answers 12 diagnostic questions evaluating problem-solving and technical affinity. A multi-attribute algorithm calculates their compatibility across 25 domains and recommends the Top 5 matches with glowing borders.

Q19: How are the 25 domain workstation images loaded?

Spoken Answer: They are curated high-definition 16:9 PNG images stored locally in img/ and mapped in js/ai.js. They load instantly in **0 milliseconds** with 100% offline reliability and zero API cost.

Q20: What is special about the 3D Resume Builder?

Spoken Answer: It features an interactive HTML5 3D vector canvas. 110 stars drift in 3D perspective space, and typing into form fields dynamically accelerates the starfield speed from 1.0 to 10.0 in real-time.

Q21: What is the Readiness Index and what is its formula?

Spoken Answer: It is a 0–100 score calculated as: (Average of best HR, Tech, and Mock scores * 0.90) + Reflection Bonus (5 points). Students must score **75% or higher** to qualify for placement certification.

Q22: When does a student receive the Placement Readiness Certificate?

Spoken Answer: When their Readiness Index reaches 75%, their status turns to 'Pending'. Faculty reviews their dossier in the Admin portal and clicks 'Approve', which unlocks their verified digital certificate.

Q23: What can faculty do in the Admin Portal?

Spoken Answer: Faculty can view institutional cohort analytics, search and filter students by department, year, or domain, inspect individual student performance dossiers, approve placement certificates, and export CSV reports.

Q24: How many APIs are used in the entire project?

Spoken Answer: Exactly **two APIs**: 1. **Google Gemini 3.6 Flash API** for AI interviews and reports, and 2. **Supabase Cloud API** for authentication and database sync.

Q25: What are your planned future enhancements for CareerForge?

Spoken Answer: Adding real-time voice-to-voice interviews using Gemini Live Audio WebSockets, automated resume PDF parsing via OCR, and direct integration with college ERP placement portals.